Exhibit

SQL computes.
The LLM explains.

Role-aware log intelligence on SereneDB

```
$ exhibit-mcp
connected: exhibit 0.3.0
12 tools

> episodes  gap 5 min
[E-55EF08]
19,250 incidents

> verify_evidence
MATCH
```

Team Logging off Early · Arnie and Adnan
SchubKI National AI Hackathon 2026 · Track 04

One question, three readers

Which problem should we fix first?

ENGINEER

Diagnose

What broke, on which machine, and what does the line say?

Raw lines · phrase search · timeline

TEAM LEAD

Assign

Whose work is it, and is it getting worse?

Signatures by owning team · weekly trend · no raw lines

MANAGER

Prioritise

What first, and why should I believe it?

Risks by blast radius and persistence · no log text

Lines are not incidents

152,183

alert lines from ONE node in 6h22m (2005-06-11): 44% of all alert lines

49,154

lines across 22,529 nodes, a fleet-wide event (2005-06-14)

348,460 alert lines → 19,250 incidents (5-min gap) → 1,377 per signature

One corpus, three rankings

Same BGL corpus, three orderings, three different first problems.

BY LINES

data TLB error

152,734 lines

117 incidents

99.64% from one node

BY INCIDENTS

Lustre + ciod

Lustre mount: 3,667

ciod socket: 3,638

at most 8 nodes in one

BY BLAST RADIUS + PERSISTENCE

rts panic

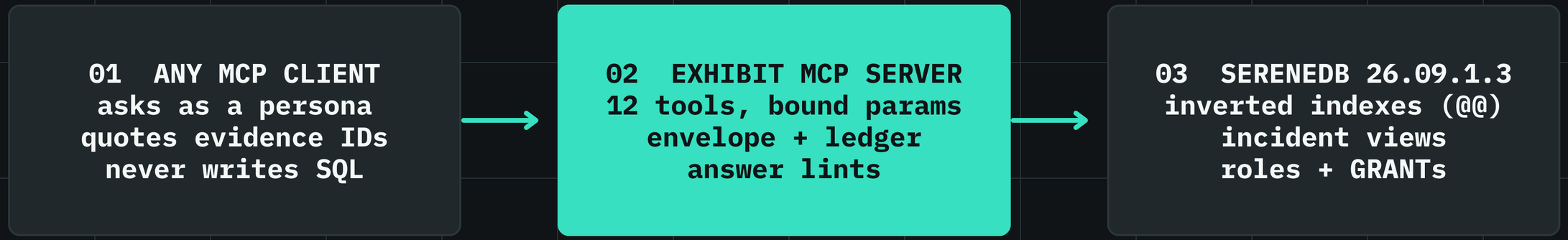
3,394 nodes

124 active days

data TLB error: 12th

How Exhibit works

The client asks. SereneDB computes. Every answer carries its query and evidence ID.



Every figure comes from SQL that ran.

Twelve tools, one envelope

ORIENT

- describe_data
- ownership

FIND

- search_logs
- nearby_events

MEASURE

- count_events
- episodes
- top_signatures
- trend
- what_changed
- durations

TRUST

- verify_evidence
- lint_answer

Every result: rows · SQL · params · true total · truncated · evidence ID

Limits: at most 200 rows per answer · 400 buckets per trend · log text quoted, never dumped · 30-second deadline, enforced client-side

Three ways to connect, one of them live

CLONE THE REPO

.mcp.json

uv run exhibit-mcp (stdio)

LOCAL HTTP

make mcp-http

127.0.0.1:8900/mcp

LIVE · ZERO SETUP

Live endpoint

streamable HTTP · read-only · same database roles

https://hermes.bushbaby-platy.ts.net:10000/mcp

Tested 18 Oct: 12 tools · E-55EF08 identical to a local run · verify_evidence: MATCH

Roles enforced by the database

`count_events(bgl, group_by=team)`: one call, three personas

ENGINEER

4,747,963

rows, counted from the raw lines

TEAM LEAD

4,747,963

lines, summed from a daily rollup

MANAGER

348,460

alert lines only, no log text

`NOT PERMITTED  lead → search_logs(bgl, "ciod")`

SereneDB refuses: an index search is privilege-checked against the base table, and the lead role has no grant on it.

W1 engineer: one machine, not the fleet

```
> search_logs(bgl, "data TLB error interrupt", mode=phrase, persona=engineer)
```



Deepest incident: R30-M0, 152,182 lines on 1 node in 6:21:47.98.

Next step: pull that node's neighbouring events (nearby_events) before touching the fleet.

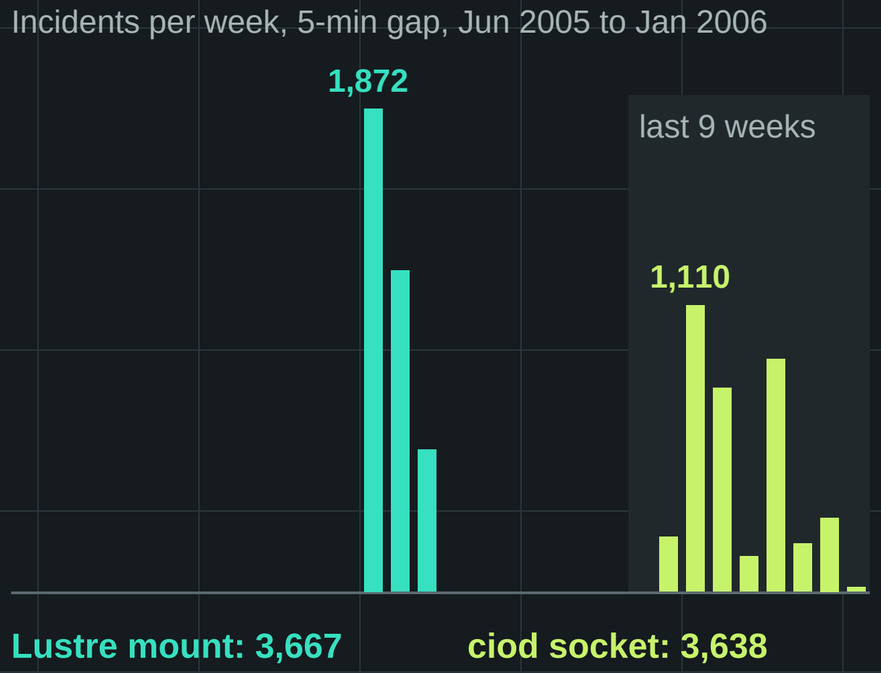
```
the gap is a parameter:
gap 1 min    132
gap 5 min    117
gap 60 min    57
deepest   E-55EF08
```

W1 team lead: one stopped, one is still active

91.1%
of all lines

51.7%
of incidents

Both shares belong to compute-platform (81.6% of alert lines). workload-runtime holds 48.0% of incidents.



Next step: assign the ciod socket error to workload-runtime now. Lustre stopped after 3 weeks; the logs do not say why.

W1 manager: a slow burn outranks a fire

348,460 alert lines → 19,250 incidents (18.1 lines each), ranked by blast radius + persistence

01 rts panic: fix first

`KERNEL · compute-platform · 3,394 nodes · 124 active days`

02 data TLB error: 12th

`1st by lines (152,734) · 541 nodes · 11 days · 99.64% on one node`

Not established by the logs: cost, downtime, impact on jobs or users, root cause.

The manager's view holds no log text and no node IDs.

Blast radius = distinct nodes affected; persistence = active days. A ranking of measured factors, not an impact estimate.

W2 OpenStack: only lifecycle finds the slow ones

ERRORS

noise

188 ERROR + CRITICAL lines: 8 tracebacks of one image-cache bug, plus 2 token incidents

LATENCY

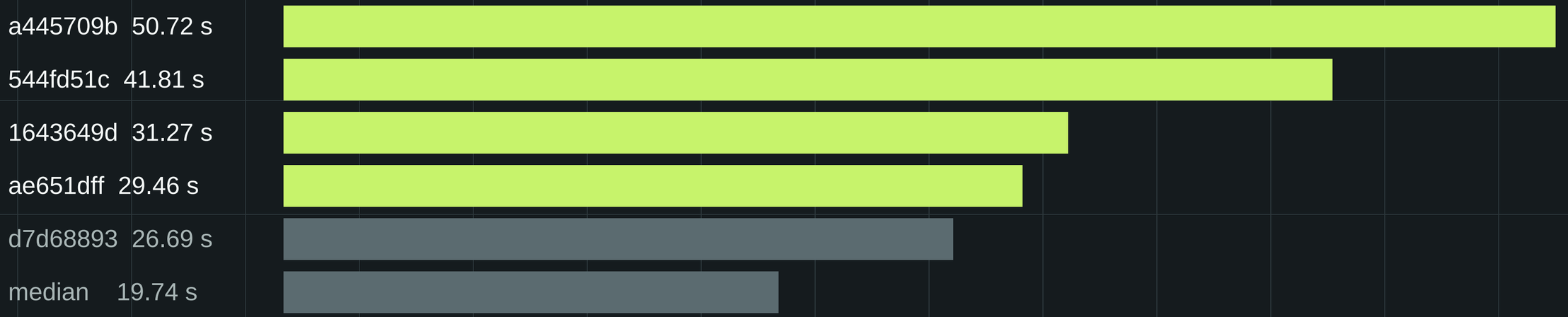
identical

API latency: 0.2340 s abnormal vs 0.2341 s normal

LIFECYCLE

separates

spawn time puts the 4 labelled instances at 1 to 4 of 2,069



Lime: the 4 instances in anomaly_labels.txt, scored by a script the agent cannot call. Error levels and latency miss them; spawn duration does not.

Every number travels with its query

```
> episodes(bgl, persona=engineer, gap_minutes=5)
{
  "evidence_id": "E-55EF08",
  "role": "exhibit_engineer",
  "total_matches": 19250,
  "returned": 1,
  "truncated": true,
  "rows": [{"signature": "data TLB error interrupt",
            "midplane": "R30-M0", "lines": 152182,
            "nodes": 1, "span": "6:21:47.98"}],
  "query": "WITH marked AS (... LAG(ts) OVER ...)"
}
> verify_evidence("E-55EF08")
MATCH: 19,250 rows recorded, 19,250 now
```

RE-RUNNABLE
the ledger keeps SQL, role and result hash

LINTED
a manager answer that claims downtime or revenue is rejected

SCORED
make check 101/101
make eval 15/15 questions
19/19 figures, 6 refusals

Scored against answer keys, outside the agent

make validate runs outside the agent. anomaly_labels.txt is read from disk; no tool, table or column holds it.

OPENSTACK
4 labelled at 1 to 4 of 2,069

BY CHANCE
1 in 761,325,852,001

BGL AGREEMENT
99.9998% of 4,747,963 lines

PRECISION
signatures 0.999991
FATAL/FAILURE 0.4065

BGL's label is part of the data the agent can read, so there we test agreement, not blindness.

Negative controls: 0 of their 100 lines above INFO;
API latency 0.2340 vs 0.2341 s. make validate: 13/13

A billion rows, searched where they sit

1B

rows of the organisers' OTel file:
41 GB on S3, 1,624 row groups,
never downloaded; its shape from
the footer in about 2 s

```
SET enable_external_file_cache = false;
CREATE VIEW bench_otel_slice AS
  SELECT ... FROM read_parquet(<S3 url>)
  WHERE "TimestampTime" in a 10 s window;
CREATE INDEX bench_otel_idx
  ON bench_otel_slice USING inverted (...);

-- 5,275,597 rows, 10 of 1,624 row groups
-- built in 60 s, 136 MB on disk
-- phrase + facet: 13 ms, 290,473 hits
```

30 s and 60 s slices crash the build on our machine: reported, not hidden. 16 vCPU, about 6 MiB/s to S3.

What we do not claim

OWNERSHIP

The team map is ours: 23 rows, fictional teams, 100% coverage.

THE BRIEF VS THE DATA

Brief: 12 categories, 5 levels. Data: 42 and 6. Organisers: the data wins.

IMPACT

No cost, downtime, customer or revenue figure: the logs cannot support one.

84,253 ASSERT lines are labelled normal; only the message text shows they are faults.

Run it yourself

Every number in this deck is printed by one of these commands.

```
git clone github.com/Arnie1x/exhibit
make up fetch parse
make bootstrap          # about 20 s
make smoke demo check eval validate
make bench              # the 1B-row file
make mcp                # or the live endpoint
```

SQL computes. The LLM explains.

github.com/Arnie1x/exhibit · MIT · Team Logging off Early, Arnie and Adnan